



Building RAG Agents with LLMs

Ragas: Automated Evaluation of Retrieval Augmented Generation

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본 자료의 저작권은 Jae Y. CHOI (jaeyeong2022@gmail.com)에게 있습니다.

본 자료는 자유롭게 이용할 수 있으나, 사용 시 저작자 및 출처를 반드시 표기해 주시기 바랍니다.

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Learning Objectives

Evaluating RAG with RAGAS

- Separate retrieval and generation errors
- Compute the three core metrics
- Interpret agreement with human judges
- Use scores to inspect failures



RAGAS

Automated Evaluation of Retrieval Augmented Generation

Diagnosis is needed for the detailed components, not the system

User Asks Something

Agent Responds

Retrieval: Tool runs algorithms (database, code execution, semantic search, return a constant value, etc) to provide context

Augmented: Based on tool responses, the software pipeline synthesizes some “context” to feed to LLM w/ question.

Generation: Based on question, instructions, and enhanced context, the LLM returns a response.

RAGAS

Automated Evaluation of Retrieval Augmented Generation



Ragas: Automated Evaluation of Retrieval Augmented Generation

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RAGAS

Automated Evaluation of Retrieval Augmented Generation

Question

q

Who directed Oppenheimer,
and who plays Oppenheimer?

Retrieved Context

c

Director: Christopher Nolan
Lead actor: Cillian Murphy

Generated Answer

a

Christopher Nolan
and Cillian Murphy.



No reference answer required!!!

Three Views of Quality

One RAG output can fail in different ways

Faithfulness

Context ↔ Answer

**Supported by
the evidence?**

Answer Relevance

Question ↔ Answer

**Addresses the
whole question?**

Context Relevance

Question ↔ Context

**Focused on
useful evidence?**

Faithfulness

Is every claim supported by the retrieved context?

1 Extract claims → **2 Check support** → **3 Count support**

Answer → claims S

Each statement + context

Yes verdicts → set V

$$F = \frac{|V|}{|S|}$$

S: all extracted claims

V: claims supported by the context

Faithfulness

Same evidence. Different support decisions.

Evidence: Christopher Nolan directed the film; Cillian Murphy plays Oppenheimer.

Supported answer

Christopher Nolan directed it.
Cillian Murphy plays
Oppenheimer.

$$F = \frac{2}{2} = 1$$

Unsupported answer

James Cameron directed it.
Tom Cruise plays Oppenheimer.

$$F = \frac{0}{2} = 0$$

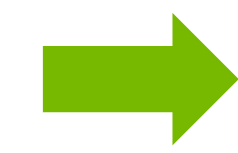
Original table → Appendix A1

Answer Relevance

Could this answer lead back to the original question?

1 Generate questions

From the answer alone



2 Compare meanings

Original q vs. generated q_i



3 Average similarity

Across n generated questions

$$AR = \frac{1}{n} \sum_{i=1}^n \text{sim}(q, q_i)$$

q: original question

q_i : a generated question

n: number of generated questions

sim: cosine similarity of embeddings

Answer Relevance

Does the answer preserve both parts of the question?

Question: Who directed Oppenheimer, and who plays Oppenheimer?

Both parts answered

Christopher Nolan and
Cillian Murphy.

POSSIBLE REGENERATED QUESTION

Who directed Oppenheimer,
and who stars as Oppenheimer?

Only one part answered

Christopher Nolan
directed it.

POSSIBLE REGENERATED QUESTION

Who directed
Oppenheimer?

Paper's example → Appendix A2

Context Relevance

How much of the retrieved context helps answer the question?

Question: director + lead actor?

Christopher Nolan directed Oppenheimer.

Cillian Murphy plays J. Robert Oppenheimer.

The film is based on American Prometheus.

It is a biographical thriller.

Emily Blunt plays Katherine Oppenheimer.

2 useful sentences out of 5

$$\text{CR} = \frac{|S_{\text{ext}}|}{|S_c|}$$

$$\text{CR} = \frac{2}{5} = 0.40$$

S_{ext} : selected sentences

S_c : all context sentences

Paper's example → Appendix A3



Building RAG Agents with LLMs

RAGAS: How Well Do the Metrics Work?

WikiEval

A small human-judged benchmark

- **50 Wikipedia pages**
- **Events since 2022**
- **2 human annotators**

Reference-free scoring still
needs human validation.

Wikipedia introduction

Context

ChatGPT

Question + answer

Human comparison

Preferred candidate

Evaluating the Metrics

Does the metric choose the same candidate as a human?

Metric

Candidate comparison

Faithfulness

With context vs. without context

Answer Relevance

Complete answer vs. incomplete answer

Context Relevance

Original context vs. extra sentences

RAGAS · GPT Score (0–10) · GPT Ranking (A/B)

Evaluation LLM: gpt-3.5-turbo-16k

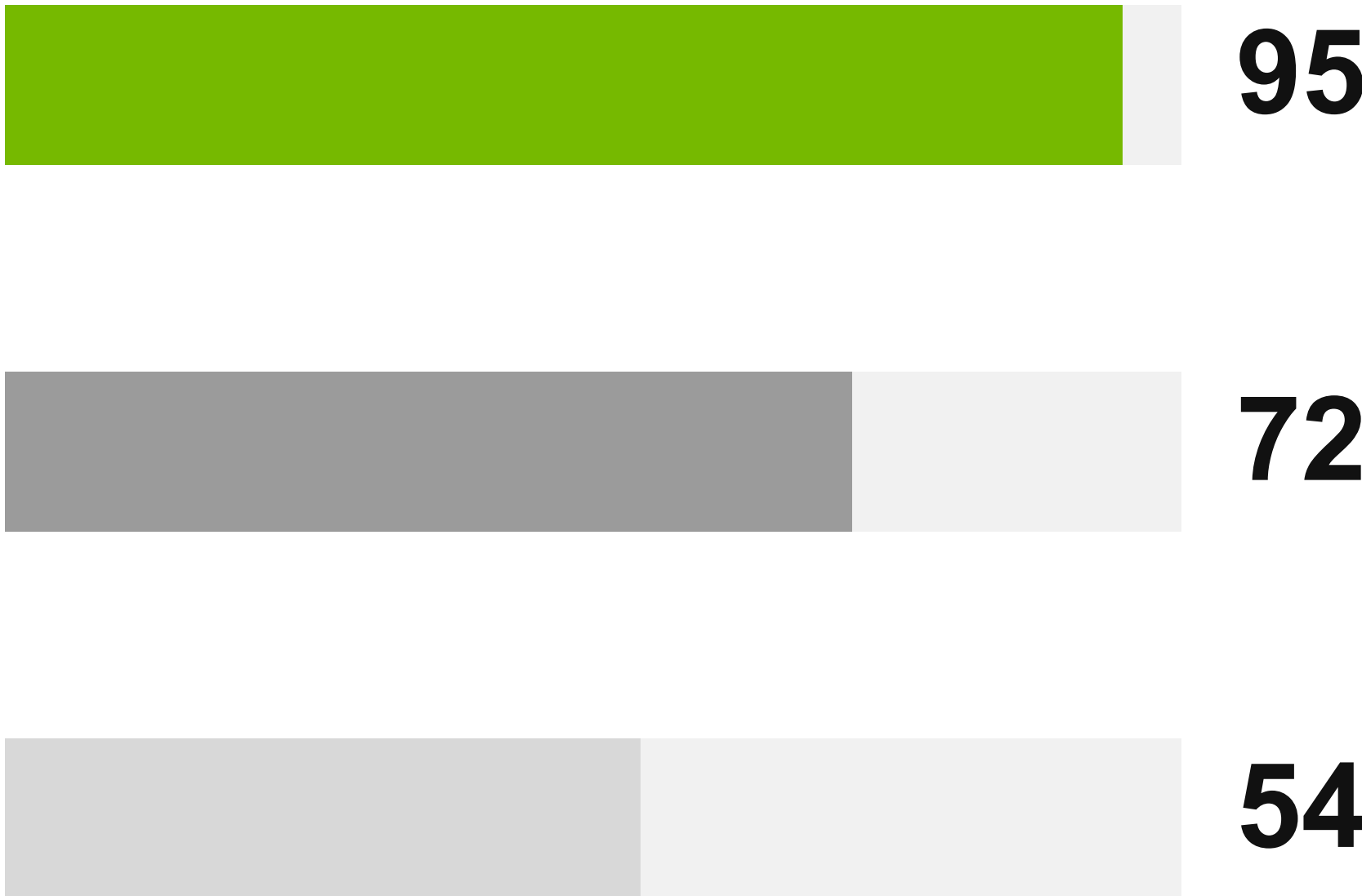
Agreement with Human Judgments

Pairwise preference accuracy on WikiEval—not RAG answer accuracy

RAGAS GPT Score GPT Ranking

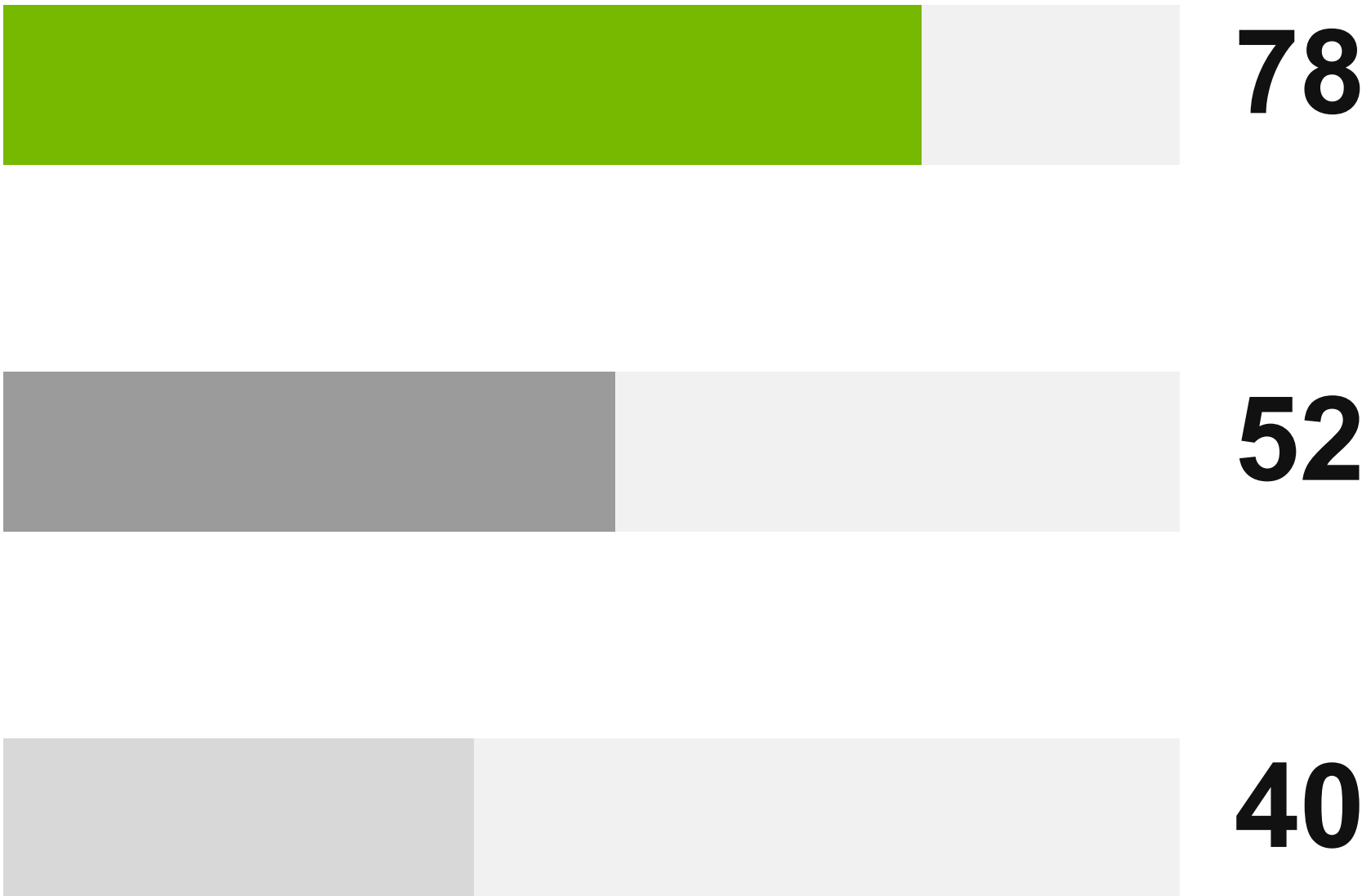
Faithfulness

0% 50% 100%



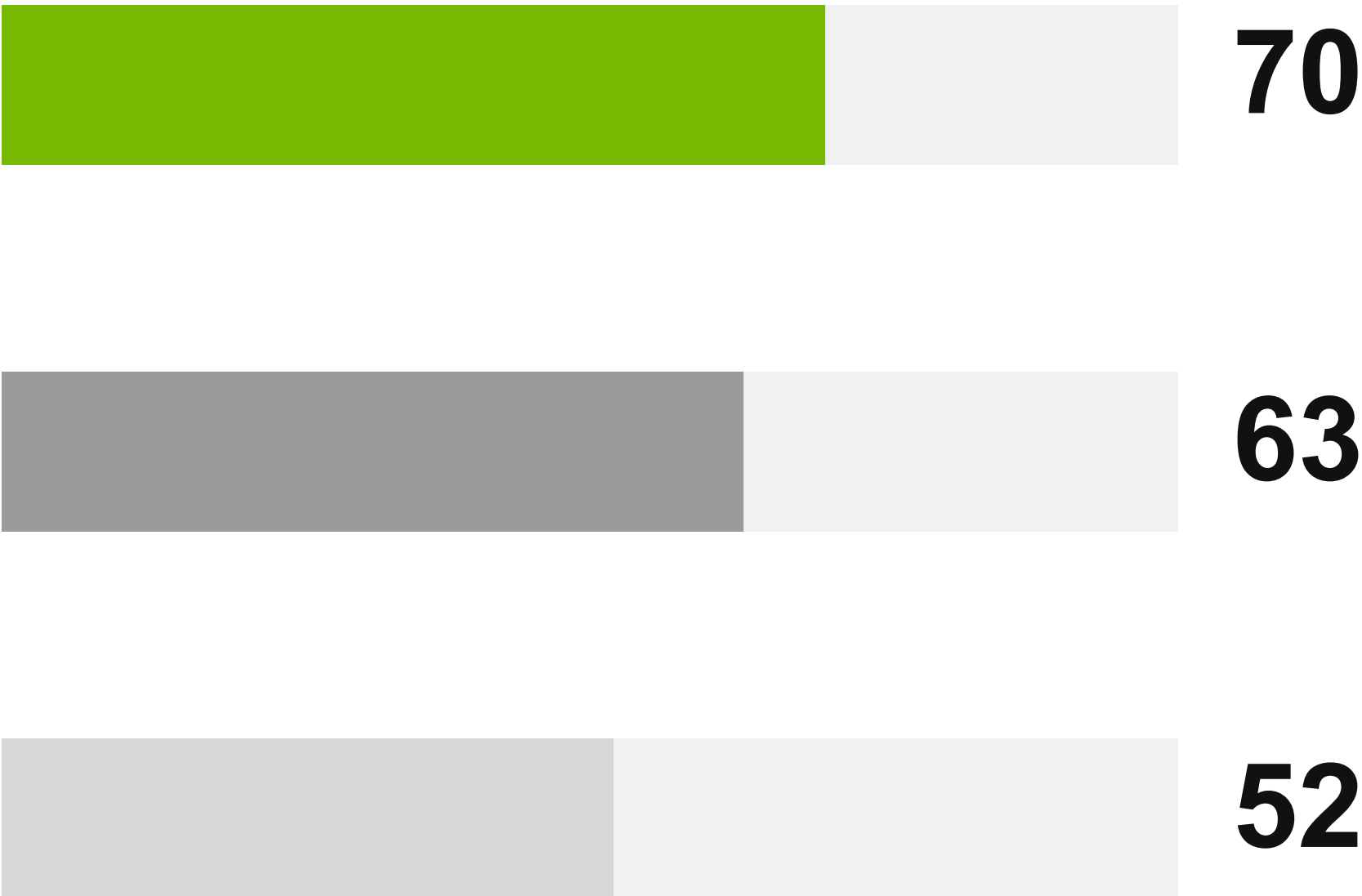
Answer Relevance

0% 50% 100%



Context Relevance

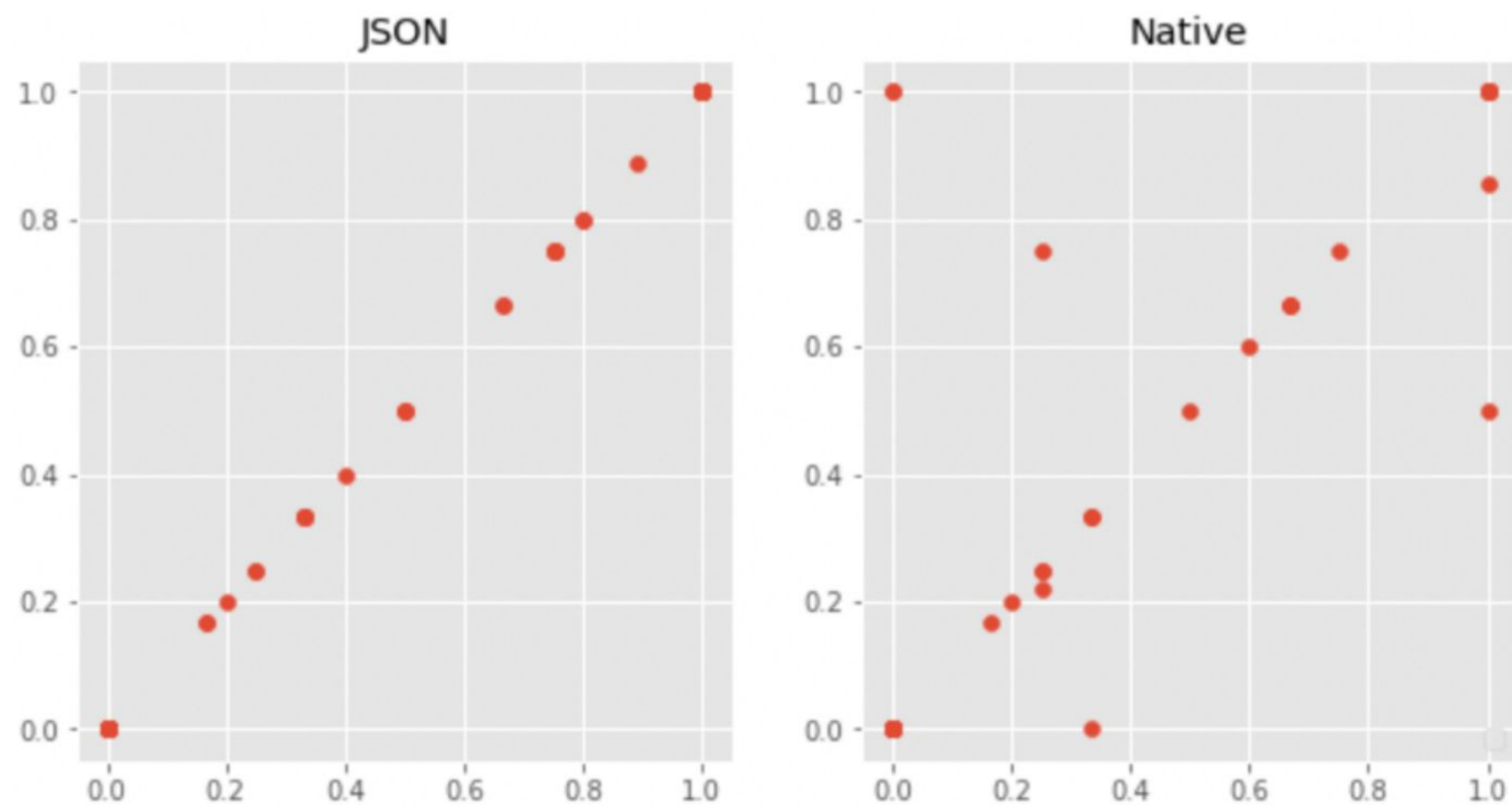
0% 50% 100%



Highest agreement: Faithfulness (95%)

Reproducibility

Structured output makes repeated scoring more consistent



Horizontal: run 1 score · Vertical: run 2 score

Closer to the diagonal

More agreement between runs

More consistent
≠ always correct

From Scores to Diagnosis

Inspect the examples behind a low score

Low Faithfulness	➡	Check unsupported claims
Low Answer Relevance	➡	Check missing intent
Low Context Relevance	➡	Check irrelevant passages

Same questions. Fixed evaluator. Human spot checks.

RAGAS: Key Takeaways

Three complementary views of RAG quality

Faithfulness

Answer Relevance

Context Relevance

Grounded?

On target?

Focused?

Use the scores to diagnose—not to certify correctness.



Building RAG Agents with LLMs

Appendix

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Appendix A1: Faithfulness

Original WikiEval example from the paper

Question	Context	Answer
Who directed the film Oppenheimer and who stars as J. Robert Oppenheimer in the film?	Oppenheimer is a 2023 biographical thriller film written and directed by Christopher Nolan. Based on the 2005 biography American Prometheus by Kai Bird and Martin J. Sherwin, the film chronicles the life of J. Robert Oppenheimer, a theoretical physicist who was pivotal in developing the first nuclear weapons as part of the Manhattan Project, and thereby ushering in the Atomic Age. Cillian Murphy stars as Oppenheimer, with Emily Blunt as Oppenheimer's wife Katherine "Kitty" Oppenheimer.	<i>High Faithfulness:</i> Christopher Nolan directed the film Oppenheimer. Cillian Murphy stars as J. Robert Oppenheimer in the film. <i>Low Faithfulness:</i> James Cameron directed the film Oppenheimer. Tom Cruise stars as J. Robert Oppenheimer in the film.

Table 1: Example from WikiEval, showing answers with high and low faithfulness.

Appendix A2: Answer Relevance

Original WikiEval example from the paper

Question	Answer
When is the scheduled launch date and time for the PSLV-C56 mission, and where will it be launched from?	<i>High answer relevance:</i> The PSLV-C56 mission is scheduled to be launched on Sunday, 30 July 2023 at 06:30 IST / 01:00 UTC. It will be launched from the Satish Dhawan Space Centre, Sriharikota, Andhra Pradesh, India. <i>Low answer relevance:</i> The scheduled launch date and time for the PSLV-C56 mission have not been provided. The PSLV-C56 mission is an important space mission for India. It aims to launch a satellite into orbit to study weather patterns.

Table 2: Example from WikiEval, showing answers with high and low answer relevance.

Appendix A3: Context Relevance

Original WikiEval example from the paper

Question	Context
When was the Chimnabai Clock Tower completed, and who was it named after?	<i>High context relevance:</i> The Chimnabai Clock Tower, also known as the Raopura Tower, is a clock tower situated in the Raopura area of Vadodara, Gujarat, India. It was completed in 1896 and named in memory of Chimnabai I (1864–1885), a queen and the first wife of Sayajirao Gaekwad III of Baroda State. <i>Low context relevance:</i> The Chimnabai Clock Tower, also known as the Raopura Tower, is a clock tower situated in the Raopura area of Vadodara, Gujarat, India. It was completed in 1896 and named in memory of Chimnabai I (1864–1885), a queen and the first wife of Sayajirao Gaekwad III of Baroda State. It was built in Indo-Saracenic architecture style. History. Chimnabai Clock Tower was built in 1896. The tower was named after Chimnabai I (1864–1885), a queen and the first wife of Sayajirao Gaekwad III of Baroda State. It was inaugurated by Mir Kamaluddin Hussainkhan, the last Nawab of Baroda. During the rule of Gaekwad, it was a stoppage for horse drawn trams. The clock tower was erected at the cost of 25,000 (equivalent to 9.2 million or USD 120,000 in 2023).

Table 3: Example from WikiEval, showing answers with high and low context relevance.



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